

Why Writing Well is Important

Today, I'm talking about how do you, as a researcher,

0:03

learn good practices about how to write?

0:07

Let's begin by talking about why you should care

0:10

about this paper's last longer than code.

0:13

So even though you may think: as a computer scientist,

0:17

I don't need to do this, I just need to be a good programmer.

0:21

That's not an accurate representation

0:24

of what an academic career is like.

0:26

Ideas are how you build your career, and publications are currency.

0:31

And while you can get a middling idea published with good writing,

0:35

the reverse isn't true.

0:36

If you have a fantastic idea, but you can't explain to anyone how it works,

0:40

it's not going to get published And eventually, once you become faculty,

0:45

you need to get money for your ideas,

0:47

and that is entirely done through writing.

0:51

You write an application for a grant, and if they like your writing, you get money.

0:56

And even if you say, Well, I'm not going into academia, I'm

1:00

going to go into industry, the same thing is true there

1:03

You need to convince the senior vice president

1:06

to allocate workers

1:08

to allocate coders to allocate computing resources to a project.

1:12

And oftentimes

1:13

you need to write up your proposal in a memo

1:15

and you need to be able to write that well.

1:18

And even if you leave research altogether,

1:21

this is a skill that translates the best in today's knowledge economy.

1:25

You need to be able to express your ideas to someone else,

1:28

and writing communication is the way to do that.

1:31

How do you get there? How do you get to being a good writer?

1:34

Norman Ramsey has seven principles of best practice and writing

1:39

will not come quickly, but good writing
1:42
is achieved through brief daily sessions.
1:45
You need to build the habit of writing
1:48
and focus on writing every day.
1:51
Don't worry about the word count.
1:54
Sometimes you just need to fail
1:57
in writing to succeed at writing the next day.
2:01
And writing is not just putting pen to paper or fingers to keyboard.
2:06
It's also about thinking about writing.
2:08
Sometimes you need to do something before you write.
2:11
You need to outline, you need to talk to people.
2:14
That's all part of the process that counts, too, and don't feel frustrated
2:18
if all you've done is write an outline.
2:19
That's often the most important part of writing.
2:22
And don't worry about bad first drafts.
2:26
A good draft is a done draft.
2:28
And don't worry about page limits.
2:30
Once you have the ideas on paper, you can cut them down,
2:33
and that is an important part of the writing process as well.
2:37
But you first need to get it on paper.
2:39
As you think about writing, you should think about writing
2:42
as documenting the ideas, not documenting the work.
2:46
One common problem that I see is that people write the paper
2:50
where the length of a section is proportional to how hard it was
2:55
to actually do the thing.
2:56
And that's not the best way to sell a reader that your ideas are correct.
3:01
You need to convince the reader
3:02
that your idea is so elegant, so beautiful that of course, it's correct.
3:06
I'm going to do that.
3:07
That's the way I'm going to think about it,
3:09
even though the first time someone does something is very difficult.
3:12
You need to show that yours is the right way and your thoughts are elegant.
3:18
And so you need to think about what's the best way

3:20
to get the reader of the paper to where you are today.
3:23
So every writer needs a reader.
3:25
How do you find readers for your paper?
3:28
So the first thing I would emphasize
3:30
is that oftentimes, especially early in the process,
3:33
the best readers are not people who know much about your subject area.
3:37
And so you can ask friends, family.
3:40
Am I explaining what's going on?
3:41
Does it make sense?
3:43
Have I convinced you that it's important people in your field
3:46
know that it's important?
3:47
You need to convince someone else that it's important,
3:49
but you only get one opportunity to get a first read.
3:53
So use this judiciously.
3:54
one resource that I think is criminally overlooked is the writing center.
3:58
And so most universities, including the University of Maryland,
4:01
have an excellent writing center where there are people who are trained
4:05
to read papers to give you feedback and to give you advice.
4:09
They're like a coach on the writing process, so take advantage of it.
4:13
They are not computer scientists usually, but they know about writing
4:17
and they know about talking about writing, and that's often very useful.
4:20
I myself was a writing center tutor.
4:23
Even when I was working there as a tutor,
4:24
I would go there and get other people's advice.
4:26
You should also be sending out your writing to competitors.
4:29
They're going to see your work eventually, and they know the work best.
4:33
And so it's better to get their feedback over the years.
4:36
You build up relationships and you get into habit
4:38
of sending each other very early drafts of your papers.
4:41
And this is a good way of learning about what's going on and to see
4:45
other people's writing process, and you can learn a lot from that as well.
4:49
And then, of course, obviously and this is the one that you probably already do
4:53

have other people in your lab.

4:54

Read what you write.

4:56

The problem here is that you need to write early.

4:58

And so this goes back to the Ramsey principles that we talked about before.

5:02

You need to start your draft early enough so that you can ask other people

5:05

to read it.

5:06

Get people who are very similar to the pool of reviewers that you're going

5:09

to get at the end when you submit your paper to a journal or a conference.

5:13

Once you get feedback, you should actually listen to the reviewers.

5:16

Let them describe the paper to you.

5:19

Don't say anything.

5:20

Let them tell you the main idea of the paper.

5:23

Treat every review like gold dust.

5:25

It's very precious and be truly grateful for the criticism, as well as the praise

5:30

that you get, and every criticism is an opportunity for improvement.

5:34

And so don't react like, no, I didn't mean that you idiot.

5:39

That means that you didn't explain something correctly,

5:42

and so fix the paper so that even those idiots

5:44

can actually understand it, like the much maligned reviewer 2.

5:48

And it's also important in reviewer responses when conference

5:53

reviews come back and you need to interact with reviewers.

5:56

And so be grateful in those author responses and reflect

6:00

your gratitude to the reviewers in those author responses.

6:03

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How to Structure Sentences and Paragraphs

Let's continue our discussion about writing.

0:03

This is often a difficult conversation to have with people in a program

0:07

because you have been in school for a long time.

0:10

You've gotten good grades in all of your classes, including

0:14

English and writing classes.

0:15

What more is there to learn?

Principles (Ramsey)

0:18

Ramsay's principle number one isn't in play here.

0:20

People usually have a good understanding of the rules

0:24

of English grammar, and that's not a problem.

0:27

But if you don't know them cold, you should.

0:30

What is more often an issue is translating what you learned

0:35

in your college and high school classes into academic writing.

0:40

Here is where sometimes you need to unlearn what you learned before.

0:44

One major problem in technical writing that often comes up is that people

0:49

have abstract ideas of the things that they're writing about.

0:52

And there are subtle nuances, and you end up with six or seven

0:57

different ideas that don't have consistent names,

1:00

and the relationships between them are explained.

1:03

You should think about this like writing a play.

1:05

If you have read a play you probably have noticed

1:07

at the beginning of the play there's a cast of characters,

1:10

and that's the way you should be thinking about writing this sort of paper.

1:13

Who are the characters that are going to show up?

1:15

What are the algorithms? What are the approaches?

1:17

What are the models?

1:18

What am I going to call them in a way that will make it

1:21

clear to the reader what their roles are, what's important?

1:25

What is my contribution here and what comes from previous work?

1:29

Once you have that cast of characters refer to them as individual actors

1:32

and use them with strong verbs so that the information flows

1:36

so that you can emphasize what's new and that the ideas

1:40
in individual units of thought are clear.
1:44
And once you have structured them, create sections
1:48
that have parallel structure so that if you use one organizational idea
1:52
in section two, that will be reflected in section three and section four
1:56
as long as it's necessary.
1:58
Oftentimes, students list and this is a good way of pre writing.
2:01
You need to list all the stuff that's important.
2:04
But then after you've listed it, you need to make it coherent.
2:08
So don't just write the list and stop
2:10
write the list and then turn it into coherent prose.
2:13
So let's dig into these a little bit deeper.
2:16
And none of this will be easy.
2:18
You need to practice.
2:19
This will not come immediately.
2:21
You need to show it to other readers
2:25
because every
2:26
writer needs a reader, and this will come eventually with time.
2:30
You also need to practice deliberate practice, make it part of your routine.

The Active Voice is Used

2:35
One thing about writing clearly is that you want to use the active voice.
2:38
Oftentimes, you have learned in high school is often---the worst offender
2:42
in the United States---that you should use a passive voice
2:46
when you're writing scholarly prose.
2:48
But I disagree with this---pretty harshly.
2:50
This often stilt papers and makes him unpleasant
2:53
to read, and it's not fun to write this way, either.
2:57
Beyond that: More importantly, take a look at this example here.
3:01
These properties were thought desirable.
3:03
Who thinks that they're useful?
3:06
Is the author of the paper.
3:08
Is it some standards body?
3:10
Is it legislation?
3:11

Say who is actually doing this?

3:13

The passive voice isn't just on fun to read and to write.

3:17

It also obscures agency and makes it harder

3:21

to understand who is doing what or why it's happening.

3:24

Another thing that you often need to unlearn using complicated language

Use Simple Language

3:29

and sometimes you want to feel important and your ideas are important.

3:34

And so we need to use complicated words for complicated ideas,

3:37

and that's usually not the right approach.

3:39

So you should indeed be as specific as possible,

3:43

but don't make things more complicated than they need to be.

3:46

You can simplify your ideas and say what you mean.

3:50

And once you figure out how to make

3:53

everything as simple as possible and as specific as possible,

3:56

you often get to this right balance between complexity and specificity.

4:01

And that is where really excellent scientific writing lies.

4:05

You do need to be specific,

4:07

but don't be overly complicated when you don't need to be.

4:11

Let's see an example of this

4:12

to demonstrate whether the proposed metrics

4:14

align with actual human experiences and examine the quality model output.

4:17

We conduct human evaluations.

4:19

So this is a lot of complicated prose with unspecific verbs.

4:23

The language is more complicated than it needs to be,

4:27

and there are verbs like conduct that mean nothing.

4:32

And so you could replace this with do we do human evaluations

4:36

so use verbs that actually convey the meaning?

4:38

So let's try to rewrite this.

4:40

Another thing

4:40

that was kind of hidden in all of this is that what are you evaluating?

4:44

Let's rewrite this to check whether the proposed metrics agree.

4:47

So let's use a more specific verb: AGREE with listeners.

4:51

We asked listeners if they can understand the lyrics

4:53
and ask singers how hard it is to sing the song.

4:55
So we're being much more specific about what we're writing about.

4:58
Say what you're actually doing.

4:59
We're asking listeners whether they like something or not.

5:03
Once you have individual sentences that make sense,

5:07
you need to order them correctly.

5:08
Let's take a look at this example where it's talking about security proofs.

5:12
Suddenly, you get the game playing technique out of nowhere.

5:16
Where did this come from?

5:17
How does this connect to cryptographic protocols?

5:20
This is totally new.

5:21
You want to move from old to new and connect the old ideas to the new ideas.

5:27
It should be clear when you're introducing something new.

5:30
And so when the reader sees something like that, the game technique?

5:35
Oh, did I miss that in a previous section?

5:36
Do I need to go back and read it?

5:38
No, it's new, but you should make it clear to the reader that it is actually new.

5:41
You can lampshade this by saying Zhang et al.

5:45
introduce THIS, we call this X because ...

5:48
but above all, explain how it fits into the context and don't hide the new stuff.

5:54
Let's see how you could rewrite this.

5:55
This is again, an example from Dreyer.

5:58
"To make it easier to manage such proofs" is how you introduce

6:02
this and say explicitly that Jones et al.

6:05
have proposed a new design principle.

6:07
Using the word proofs explicitly connects it to the previous sentence

6:12
that is talking about security proofs of cryptographic protocols.

6:15
And then you introduce a new design principle called the game playing

6:19
technique.

6:19
Then you link the next sentence to the thing that you just introduced.

6:23
This technique follows yada yada.

6:26
You explicitly made it clear

6:27

that you're introducing something new and connecting it to the old stuff.

6:30

And then as you carry on with the new stuff,

6:33

it's clear how this connects to what you've just introduced.

Flow is not enough

6:36

But flow is not enough.

6:37

So here's another example from Dryer: a perfectly lovely paragraph

6:42

with good flow between the individual points.

6:46

But what is this paragraph about?

6:48

You get to the end, and then there's this whether house Cats

6:52

can be trained to be more sociable.

6:54

That's kind of dancing around the main idea here.

6:56

And what is the main idea that needs to be at the very beginning?

7:01

It aids the skimming.

7:03

This would be the first sentence of the paragraph

7:05

and helps to organize your own thoughts.

7:06

So like one way to write

7:07

a section is to go through, write the topic sentence of each paragraph,

7:11

then fill in all of the paragraph in that section.

7:14

The topic sentence of this paragraph should actually be there appears

Adding a Topic Sentence

7:18

to be a negative correlation

7:19

between the charisma of a species and its ability to survive.

7:22

That's the main idea.

7:23

Then you can get the details about how this relationship plays out

7:27

with lions and house cats.

7:31

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7:59

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What a Good Paper Looks Like

Let's continue our discussion about writing by talking about

0:03
what does a good paper look like?

0:06
first, let's begin with the simple stuff: follow the rules.

0:09
You should submit your paper by the deadline.

0:11
Many groups have internal deadlines.

0:13
These are just as important as external deadlines

0:15
and follow the rules of whatever venue you're submitting to.

0:19
Look at the call for papers, and they usually have a sample submission.

0:23
Read that carefully.

0:24
It's not just a document that you delete all of their stuff and put your stuff in.

0:29
The stuff

0:30
that was there in that sample document tells you how to format your document.

0:33
Follow those rules.

0:34
If you submit to a place long enough, you will learn and internalize those rules.

0:38
But your first couple of times: look at it.

0:41
Read through it.

0:42
It will tell you: Here's how you should format a caption.

0:46
Here's how you should format a table. Follow those rules.

0:48
Many of your simple questions will be answered there.

0:51
Don't tweak the layout if you're short on space.

0:53
Edit the text.

0:54
Don't do `\vspace{}` hacks that can get your paper desk rejected.

0:58
Sometimes you'll squeak by, but it's a bad habit to get into.

1:02
And you should focus on editing the text rather than trying to cheat the layout.

1:06
Many of us work on software where we have things

1:08
like continuous integration for our compilation.

1:11
You should use the same process for writing your paper!

1:14
Have processes that check whether all of your references are defined,

1:19
whether you have spelling errors.

1:21

Our group has a style checker.

1:23

Niel Spring from the University of Maryland has created a number of tools

1:29

Ruby Scripts that help you check whether your paper has style errors or not

1:33

that talk about many of the rules that we talked about before.

1:37

So you can run these as continuous integration scripts check to make sure

1:40

that everything is going OK, especially when you get near the deadline

1:43

and say you have five people writing a paper.

1:45

Errors pop up!

1:46

If you have

1:47

these tools that can catch those errors, you can save yourself embarrassment.

1:51

Make sure that your paper looks good.

1:52

Make a good first impression.

1:54

Many people will skim a paper

1:55

before reading it, and so like they open up the tab on ArXiv ...

2:00

They see your paper, and I am I going to read this or not?

2:03

Oh, this looks kind of messy.

2:05

And then maybe the download it. Never read it.

2:08

Maybe they just close the tab without looking further.

2:10

You only have one chance to make a good first impression,

2:12

so make sure that that things look good. Aid skimming. So

2:15

maybe someone will read your

2:17

paper at first, they'll just skim it, but make it easier for them to skim.

2:21

Make all the figure self-contained.

2:23

Every caption should have a takeaway. Why is this figure here?

2:25

What is it telling the reader?

2:27

Don't just say summary of our result.

2:28

Talk about why your results are important: to test whether our fancy pants

2:33

algorithm succeeds at better segmenting Chinese text.

2:38

We evaluate blah blah blah.

2:39

Make the figures consistent, readable and attractive.

2:42

It should be obvious that these figures belong in the same paper.

2:46

Don't generate one figure in excel.

2:48

one figure on Google Sheets and another figure in ggplot2.

2:52
They should look like they belong together.

2:54
Make them attractive.

2:55
People often create figures and then scrunch them down at the last minute

2:59
so that they're barely readable and you need to zoom in on the PDF.

3:02
Don't do that.

3:03
Color is good, but at the same time, it shouldn't be necessary

3:06
for understanding the figure.

3:08
Some people are color blind.

3:09
Some people review papers with black and white printers.

3:11
Use color, but don't depend on it.

3:14
one kind of controversial piece of advice that I would like to give

3:17
is that the related work should come later than a lot of people do.

3:24
And here's a typical organization for a standard computer science conference

3:28
paper.

3:29
The related work shouldn't come at the beginning of the paper.

3:32
The related work shouldn't stand between your idea and the reader.

3:36
Someone reading the paper has budgeted

3:39
a certain amount of time to skim the paper to get to the idea.

3:42
If you bog them down in related work, it will take longer for them

3:45
to get to your idea, and that's the real point of a paper in the first place.

3:49
What you can talk about is the essential prior work

3:54
that is necessary for understanding your idea and getting credit.

3:57
And so this goes back to the citation rules that we talked about before:

4:00
cite generously, cite the things that are relevant, but talk about how

4:05
that specifically relates to your idea and do the bare minimum first.

4:10
Later, explicitly

4:11
write the related work about all of the things that are relevant.

4:14
You should also have evidence and interconnection between the ideas

4:18
in your paper.

4:19
The introduction will make claims: check that each claim is supported

4:22
by evidence later in the paper in the results section.

4:25
This is a good opportunity for forward thinking,

4:28

and so you can talk about all the claims that you make
4:31
in the introduction and point to the relevant sections that do this.
4:34
And this is a lot better than doing things like Section two introduces the problem.
4:39
Section three talks about our methods.
4:40
Section four evaluates our method.
4:43
Finally, Section eight concludes that could be in any paper.
4:46
Any sentence that could be in more than five papers doesn't belong in your paper.
4:50
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How to Write an Effective Abstract

We've been talking about writing, and we've been talking
0:03
about pretty big picture ideas and nuts and bolts stuff,
0:07
but let's talk about some of the important building blocks
0:10
of an academic paper, particularly writing the abstract and introduction.
0:15
I want to specifically emphasize this because the abstract is often
0:19
the first thing that people see
0:21
of your paper, and sometimes it's the only thing that people see.
0:25
Because they read the abstract,
0:26
they're bidding for papers that they're going to review, or
0:30
they're just deciding whether this paper is relevant or not.
0:33
This is kind of the make or break point to make sure that during paper review,
0:37
you get the right reviewers and later, whether people actually read
0:41
and eventually cite and use your paper ideas.
0:44
Building some ideas again from Derek Dreyer ...

0:46
He proposes a CGI model of writing an abstract
0:50
CGI stands for context, gap and innovation.
0:55
So first, what is the context?
0:57
What are you writing about?
0:59
And so let's say that we're trying to write an abstract
1:01
for everything that we're talking about in the series of vignettes.
1:04
The context writing is important.
1:07
Despite the importance, it's hard to do it well.
1:09
There are lots of people who don't know how to do it well.
1:12
And so we're going to talk about how to become a better writer.
1:16
The innovation is the part where you sell the paper.
1:18
The context and the gap are where you position your paper within your field.
1:23
Another problem that I see is that oftentimes
1:26
the introduction just repeats the abstract,
1:28
but the introduction is different from the abstract.
1:30
The abstract needs to attract the right reviewers.
1:33
But the introduction is where once you have reviewers,
1:36
you need to start them on the intellectual journey
1:41
to understanding and appreciating your idea.
1:43
It needs to build interest in the non target audience,
1:47
and it should have specific examples that help explain
1:52
why your idea is important, why it makes sense , and why it's reasonable.
1:56
You don't have space to do this in the abstract.
1:58
The abstract is pretty abstract.
2:01
The introduction can and should be much more concrete
2:04
and motivating example is usually a good idea in the introduction.
2:09
If you can figure out how to do that, then it explains a logical structure
2:13
of the paper. We'll talk about how to do that well in a bit.
2:15
The introduction also can connect the paper more broadly.
2:18
The abstract has the context.
2:21
It's just one sentence.
2:21
The introduction can connect ideas in other fields.
2:26

Contrast with other work.

2:27

You can do that much more explicitly than in the abstract.

2:30

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Citation Mistakes

I want to talk specifically about citation, because this is something

0:04

that often doesn't get taught in a standard undergraduate curriculum.

0:10

How do you cite in a scientific discourse?

0:13

You learn about citation for quotes and literature and whatnot,

0:17

but if you haven't taken a technical

0:18

writing course, people often don't know the norms of citation.

0:22

So I want to talk about this kind of nuts and bolts approach first,

0:25

and then we'll talk about how this fits into related work in a little bit.

0:29

one problem that I've been seeing a lot---and ArXiv

0:32

hasn't helped with this---is that people are only citing recent stuff.

0:35

Ideas are often pretty old.

0:37

So you should go back to the first person to actually think of the idea,

0:43

think of the problem.

0:44

And usually that's more than five years old.

0:46

At a higher level, you should cite others as you would wish them to cite you

0:50

the golden rule of citation.

0:52

So don't criticize, be generous.

0:54

Don't cherry pick parts of a paper, represent papers

0:57

in their totality, because that is how you would like to be cited, right?

1:01
As we mentioned before, don't write for length.
1:04
There are usually page limits and conference
1:07
submissions and journal submissions.
1:09
Don't worry about that yet.
1:11
Write for your understanding.
1:12
first, write down everything that's relevant.
1:14
Cite all the relevant work, build the historical narrative and then you can cut.
1:19
This is going back to the Ramsey principles that we talked about before,
1:23
and I don't want to pick on ArXiv.
1:25
ArXiv is great.
1:26
It's a great resource, but it's a preprint server.
1:29
These things have not been reviewed, and oftentimes there is good work on ArXiv.
1:33
That good work eventually gets turned into a peer reviewed publication,
1:37
so cite that instead of ArXiv, if at all possible.
1:40
And this should be obvious, but I have seen this go wrong.
1:45
Read the things that you cite.
1:46
Sometimes people cite something because everybody cites it
1:50
that isn't reason enough for you to cite it.
1:52
You should actually read it.
1:54
This may involve going to a library.
1:55
You may need to do some work to actually read the thing,
1:58
but you should read it if everybody's hiding it.
2:01
It's important.
2:01
Probably you should see what's in there so that you know whether you should
2:05
actually cite it, not just cite it because everybody else does.
2:08
That's why you should cite and when you should cite.
2:11
But how do you cite?
2:13
Citations are a little like a newspaper headline.
2:15
After you see a good citation, you should know for sure
2:20
whether you should read that or not, and you should have a good sense
2:23
of what's in the paper.
2:24
So here's a sentence we analyze the messages
2:26

through persuasion tactics which have been used in cybersecurity.
2:29
This is too broad.
2:30
Just knowing that something
2:31
has been used in cybersecurity doesn't tell you how it was used
2:35
or why it's relevant to the text messages that we're talking about here.
2:39
Let's rewrite this.
2:40
So "we analyze messages using persuasion tactics, which oliviera at all
2:45
apply to detect spear phishing emails and anon apply to persuasive blogs".
2:50
We've added additional citations here to flesh it out, to connect to more things
2:54
and to show why the Oliviera citation is actually relevant
2:58
to the text based problem that we're talking about here.
3:01
And so now you have a good sense of what's going on and what tactics they use.
3:05
And so now you have a good sense of whether
3:06
you should actually read that we cite a lot, so make it look nice.
3:10
So don't have huge citations dumps where you cite tons of things!
3:14
Either there's a representative out of there
3:15
that you should be citing instead.
3:17
And you can do inter alia, which means among others.
3:20
Or you should expand that out more and to show the contribution
3:23
of the individual work.
3:24
Citations should either be a standalone noun by themselves
3:28
, or it should be attached to a noun as a net positive.
3:32
If a citation introduces an abbreviation,
3:35
you should make that part of the citation, so it should be inside
3:38
the parentheses or the brackets where you have the citation.
3:41
That makes it look nicer.
3:43
Rather than having two sets of parentheses or brackets right next to each other.
3:46
Citations go outside of quotes, but inside the sentence,
3:49
particularly for LaTeX which a lot of us use for our citations
3:52
in computer science, make sure that things get capitalized correctly.
3:55
If things are a proper noun like the surname Dirichlet, that needs
3:59
to have a curly bracket around it so it doesn't get mangled.
4:02
You might say, why is this important?

4:03
It shows that you take care in your writing,
4:06
and if you're sloppy in your bibliography, maybe you were also sloppy
4:11
in collecting your results,
4:13
and so you should have a good attention to detail everywhere.
4:16
This is just a single lecture from a course.
4:18
YouTube likes to show you
4:19
these videos out of order, but if you go to the course web page
4:22
linked below, you can see the lectures in the right order
4:26
and you can get resources like home works or suggested reading.
4:30
You can also visit QANTA dot org if you want to learn
4:35
about our systems for creating computers that can answer questions.
4:40
Where QANTA stands for question answering is not a trivial activity.
4:44
If you want to help the channel provide a big gradient to the algorithm by

Writing in Latex

so today we're going to talk about
0:02
papers and writing and this isn't a
0:06
usual topic in a graduate course but I
0:11
realize that it isn't taught in a lot of
0:14
places and it's just kind of stuff
0:15
you're supposed to know and I want my
0:19
students who take this course to know
0:21
this sort of thing and because this is a
0:24
course about language and how computers
0:29
can understand language I think that
0:31
it's particularly embarrassing if we as
0:35
people working on computational
0:37
linguistics and natural language
0:38
processing can't string a couple of
0:40
words together to create a coherent
0:42
papers so I think it fits in this course
0:45
and what I'm going to talk about is how

Overview

0:49
you can be an effective writer in the
0:53
area of computational linguistics and
0:55
natural language processing and I'm
0:57
going to split this into three parts so
1:00
first I want to talk about the mechanics
1:02
of writing a scholarly paper in our
1:06
field and this is going to be a lot of
1:09
nuts and bolts stuff but these are a lot
1:12
of things that I see students not
1:16
learning or getting wrong quite often
1:18
and I want to talk about that just so
1:21
that you can be aware of it and
1:23
hopefully have a leg up and learn about
1:26
more important stuff and I also want to
1:28
talk about the importance of writing
1:30
well focusing on writing to the
1:33
different audiences that can read the
1:35
sorts of things that you're writing and
1:38
finally I'm not going to talk about
1:40
writing per se but what it means to be
1:44
an effective reviewer and understanding
1:47
the process of reviewing is good for you
1:49
writing a paper in that you can
1:51
hopefully have a higher probability of
1:53
having your paper accepted but also so
1:56
that you can be an effective reviewer
1:57
and help other people create
2:00
well-crafted ideas and I want to add a
2:04
disclaimer a lot of what I'm going to
2:06
say today is my pain
2:07
and that doesn't mean that I'm
2:10

necessarily right and you can definitely

2:12

disagree with me and I'd love to have

2:15

that discussion so the first thing that

2:16

I want to say is that when you write a

2:18

paper you should use law tech and low

2:22

tech was created by Don Knuth and

Why use LaTeX

2:25

emeritus professor at Stanford

2:26

University who wanted to write a series

2:31

of books about computer science and

2:33

realized that he couldn't find the software

2:35

to write those books didn't exist and so

2:37

he had to make it himself

2:39

and that was tech that was later

2:43

extended into law Tech which is the

2:46

de-facto method of creating scholarly

2:49

publications and the reason that you

2:52

should use LaTeX is that it's very

2:55

effective for creating equations you can

2:58

typeset math very beautifully it also

3:01

saves you time most journals and most

3:05

conferences provide templates in LaTeX

3:08

tech and as usually happened your paper

3:12

will get rejected from one conference

3:14

you need to reformat it for a different

3:15

conference and that's basically just

3:18

swapping out these template files and

3:20

you can focus on improving your writing

3:24

and not spend a bunch of time fiddling

3:27

with formatting making sure the headings

3:29

look right and everything else LaTeX

3:31

tech abstracts that out so that you

3:33

don't have to worry about it a pragmatic

3:36
reason to use la tech is that it's free
3:38
it's open source anyone can use the code
3:41
anyone can extend it you don't have to
3:43
pay to use law tech interpreters or
3:46
compilers and this is important
3:51
especially for people in the developing
3:53
world where their universities don't
3:55
provide them with word for free and this
4:01
allows you to collaborate more freely
4:04
with everyone in the world and that I
4:06
think is important goal of science and
4:11
another egalitarian reason that you
4:14
should prefer to use wha tech is that
4:17
for blind people it's much easier for
4:20
them to
4:21
you read la tech source code then to
4:23
read a PDF or in Microsoft Word document
4:26
la tech encodes mathematics in a way
4:29
that screen readers can deal with and
4:33
the blind students that I've worked with
4:35
much prefer having the law tech source
4:39
than looking at a PDF or looking at a
4:41
Word document all that being said this
4:44
is not a law tech tutorial most people
4:46
figure out how to use law tech just fine
4:49
they can produce documents what I'm
4:51
going to talk about are some of the
4:52
common errors even once you figure out
4:55
how to do the basics of law tech that
4:58
persist and make your documents look a
5:00
little less professional than people who
5:02
have been using law tech for a long time
5:05

and know kind of the silly little
5:07
secrets that give away how experienced
5:10
you are with law tech and although law
5:13
tech is great it's not very user
5:15
friendly and so you need to be
5:18
apprenticed to some law tech master to
5:20
learn a lot of these tips and I'm not
5:22
going to talk about all of them today
5:24
there are a lot more on my web page
5:25
where I compile a list of stuff for
5:28
students to use as resources as they're
5:32
becoming more proficient at writing so

Common mistakes

5:34
what are some of the common mistakes
5:36
that people make in law tech so first
5:39
many people position footnotes
5:41
incorrectly and so one of the great
5:43
things about law tech is that you can
5:45
very easily insert footnotes and
5:47
footnotes always go after punctuation at
5:50
the end of the sentence and so you
5:51
shouldn't have the period of a sentence
5:56
after the footnote another problem that
5:59
people have is using dashes hyphens and
6:04
in dashes incorrectly so if you have two
6:08
words if you have two words that you're
6:11
joining together with a hyphen
6:13
you do it like so one - you hit the -
6:17
button on your keyboard once and
6:19
similarly if you're typing my name boyd
6:22
Graber one - in there if you're giving a
6:26
range of numbers you have two hyphens
6:30
and so between the years 1980

6:34
1983 two hyphens pages 78 to 101 two
6:40
hyphens and also notice that just like a
6:43
- connecting two words together if you
6:45
have an in - this is called because it's
6:49
the same width as an in you don't have
6:53
spaces around that and if you're using
6:55
an in - here called an in - because it's
7:01
the same width as an M which is used to
7:05
set off losses which it never is that
7:10
also has no spaces around the - but it
7:13
has three hyphens so this is something
7:15
that many people get wrong it just
7:18
requires you to remember the difference
7:20
between hyphens in dashes and in dashes
7:24
another thing that people often get
7:26
wrong in law tech are quotes and so the
7:29
left and right quotes in law tech are
7:31
differentiated and you don't use the
7:35
same character for both you have a two
7:39
characters you type the keyboard twice
7:40
to get the left quotes and the right
7:44
quotes and on us keyboards the single
7:51
left quote is at the top left of the
7:54
keyboard and the single right quote is
7:58
in the middle on the right hand side of
8:00
the keyboard so those are two separate
8:03
characters and the two directions are
8:07
different if you use the standard quote
8:10
which is a single character it will
8:12
look okay but it won't be right in wall
8:17
tech another problem that I often see in
8:20
law Tech is in math and so you need to
8:25

be careful when you're typing
8:27
mathematical equations some things
8:29
require space around them so for example
8:32
the given sign in a probability
8:35
shouldn't just be the pipe you need to
8:39
put space on the left and the right of
8:41
it so that it doesn't look squished and
8:44
likewise logic has
8:47
nice parentheses that get bigger or
8:50
smaller depending on the stuff inside of
8:52
it and it's important to use these
8:55
correctly so you have left parentheses
8:59
and so this is one command and right
9:02
parentheses and if you just use the
9:05
single character for the left or right
9:07
parentheses it will look tiny and not
9:10
match a big expression like what's
9:14
inside here like I said before I have a
9:17
lot more of these things on my silo cage
9:19
but these are the most common you can
9:20
check out those links and see what other
9:25
la tech tips I have their images and

Images and figures

9:28
figures are also very important and I'll
9:31
talk more about why they're important
9:33
from sort of a pragmatic perspective but
9:36
when you create a figure whenever
9:39
possible use vector graphics vector
9:44
graphics are when you describe an image
9:46
based on curves and equations you say I
9:51
have a line that tilts this kind of
9:54
angle and so that that's over here if
9:57
you're drawing an e using Bezier curves

10:00
and things like that
10:02
raster images are when you divide an
10:05
image into pixels and you describe what
10:08
pixels have what colors and this looks
10:13
okay if your image isn't too big but if
10:16
you zoom in you start to see the grid or
10:19
the raster that's why it's called a
10:21
raster graphics and then if it gets too
10:26
big the image doesn't look so great and
10:29
the thing about documents is that many
10:34
times people will zoom in they'll want
10:37
to look at your figures more carefully
10:39
or they'll want to put it in a slide and
10:42
if you don't use vector graphics when
10:45
they zoom in
10:46
look that and sometimes when you go off
10:50
and print the document even if it looks
10:53
okay on the screen sometimes raster
10:55
graphics will look really bad on a
10:56
printer so it helps you look a little
10:59
bit more professional to use vector
11:02
graphics whenever possible and so use
11:06
programs like illustrator OmniGraffle or
11:13
Inkscape to create vector graphics that
11:18
you can insert in your documents and
11:20
create graphs whenever possible
11:23
they're usually easier to interpret than
11:26
tables you get a quick impression and
11:29
it's it's more efficient whenever you
11:34
have figures whether it be table or
11:36
graphics or charts use effective
11:41
captions explain what's going on what's
11:44

the takeaway of this figure don't just
11:46
say this is accuracy versus time explain
11:49
why the reader should care about that
11:51
and in your tables don't use vertical
11:53
lines only use horizontal lines and use
11:57
them sparingly again I have more about
12:00
this on my style page so I've been

Collaboration

12:03
talking about using Ia tech and one of
12:06
the corollaries of using Ia tech is that
12:10
often you'll create these logic
12:12
documents collaboratively you'll be
12:14
working with a team and how do you
12:16
manage that when I work elaborately on a
12:21
lot X paper I like to use version
12:24
control and as you use version control
12:28
you should do the following things
12:30
version control has a lot of nice
12:32
features it helps you collaborate even
12:34
asynchronously so something like Dropbox
12:36
doesn't let you do that it gives you an
12:40
implicit backup so all of your stuff
12:42
remains backed up and you can go back in
12:44
time what was I doing on a Saturday
12:48
let's go back to that version and it's
12:50
not relying on the cloud you can work on
12:52
airplanes
12:52
the internet goes down people can still
12:54
keep working but if you're using version
12:56
control you have to do the following
12:59
things to make the collaboration easy
13:01
you need to split a big document into
13:04
smaller documents and logic makes it

13:06
easy to do that with the input command
13:08
so the input command you can give
13:11
another file name and that just inserts
13:13
that other law tech file into this big
13:15
document so I like to have each of the
13:18
subsections in its own file so that
13:22
people can say hey I'm going to work on
13:23
this subsection let me take the token as
13:27
it were and only and be the only person
13:31
editing that document one of the nice
13:33
things about version control is that if
13:35
people accidentally modify the same file
13:37
it will do as good of a job as it can
13:40
merging those changes together but to
13:44
make sure that that happens in a sane
13:47
way you can't have basically all of your
13:51
text on a single line version control
13:53
merges individual lines and if only one
13:55
person has merged each line no problem
13:58
but if multiple people have edited the
14:01
same line it will know how to merge them
14:03
together and this results in a conflict
14:06
and so the more lines you have the lower
14:09
the probability that if two people
14:11
accidentally edited the same file then
14:13
you'll have a conflict so you should
14:16
flow your lines so that for example
14:20
after every sentence you start a new
14:22
line in your file and I also like to not
14:25
have any lines longer than 80 characters
14:28
so how do you keep track of who's
14:30
editing what files I like to use Google
14:31

Docs to do that and so you write down
14:33
all of the files in your paper and then
14:37
each person can write their name next to
14:40
the file that they're editing and once
14:42
they're done editing they commit their
14:44
changes and in version control and then
14:46
other people can view those changes and
14:48
edit those files after they've released
14:50
their token as it were so one question
14:53
that I often get is why not use
14:55
something like overleaf or share law
14:57
tech I've had some bad experiences the
14:59
sites are not perfect sometimes they go
15:01
down
15:01
when they go down especially when a lot
15:04
of people are using it right before a
15:05
paper deadline and that's bad news
15:08
because you don't have your own local
15:09
backups and something like version
15:13
control you have your own local
15:15
repository especially for things like it
15:17
you can keep working even if the
15:18
internet or one of these sites goes down
15:21
and one day maybe they'll get good
15:23
enough that I'll trust them again but hi
15:25
not yet so today we've talked about the
15:28
mechanics of writing this can be a
15:31
little daunting especially if you
15:32
haven't used law tech a lot but don't
15:35
let that hold you up the most important
15:36
thing is to write and it doesn't matter
15:38
if you're writing in law tech or word or
15:40
with pen and paper the ideas and getting

15:43
them in written form is the most
15:46
important thing so if psychologically la
15:48
tech intimidates you that isn't a
15:50
problem right however makes sense to you
15:53
and it doesn't matter where how the
15:57
important thing is to get your thoughts
15:58
out that's the first step and then later
16:01
you can worry about all this annoying
16:03
law tech business

WRITING for multiple audiences

in the last video I talked about the
0:01
mechanics of writing and now I want to
0:03
talk a little bit more about the actual
0:06
craft of writing and this is not a
0:10
writing course I can't do justice to uh
0:15
the many many pages that have been
0:17
written about what it means
0:19
to uh write well and so here are two
0:23
books that I really like uh style
0:25
towards Clarity and Grace and on writing
0:27
well uh they talk a lot about about very
0:30
important stuff even to computer
0:32
scientists uh things like uh structuring
0:36
your sentence to go from the familiar to
0:38
the unknown to Define terms well uh
0:43
to uh avoid vag words to use topic
0:47
sentences for every paragraph to remove
0:51
fluff and obfuscation from sentences
0:55
these are really great books I I
0:57
encourage you to uh read them on a
1:00
regular basis because every time you
1:03

read them you discover something new and
1:04
and become a better writer but I cannot
1:09
uh do justice to the ideas in these
1:11
books they really stand on their own
1:13
please do uh check them out uh instead
1:16
I'm going to talk about something a
1:17
little bit more specific to writing
1:18
papers and computational Linguistics and
1:20
natural language processing the kinds of
1:23
audiences that you'll be writing for and
1:27
these books are are more General about
1:29
writing
1:30
non-fiction and writing scientific
1:33
papers is a little bit different because
1:34
the audiences
1:37
are uh more nuanced and so I I want to
1:40
focus on that but again that's not the
1:44
only important thing about writing uh
1:46
check out these books uh lots of great
1:48
lessons in there but one of the things
1:50
that these books will tell you is that
1:52
being a good writer involves paying
1:54
attention to your readers and scientific
1:58
writing has a lot of different readers
2:01
there are the reviewers who are deciding
2:03
whether to accept your paper there are
2:05
struggling students trying to replicate
2:07
your paper uh there are journalists
2:10
trying to write a article under a
2:12
deadline and there's their friends and
2:15
family uh from uh high school or your
2:19
hometown uh checking out your web page
2:21
and seeing uh what is Sally up to once

2:25
she went off to
2:26
University and a common failure mode of
2:30
papers is focusing on one aspect of the
2:33
audience and ignoring uh everybody else
2:36
often times you just write for the
2:37
reviewers and and that can be a
2:40
problem and for example if you focus on
2:43
the experts you'll confuse uh struggling
2:48
students or the general public and if
2:51
you focus on the general public then you
2:53
can omit the details necessary for
2:56
someone to replicate your work and and
2:58
that destroys the scientific value of
3:01
your paper
3:03
and prevent someone from thoroughly
3:05
reviewing your work so what I want to do
3:08
is talk about each of these audiences in
3:10
general and what you can do to make your
3:12
paper work for each of those audiences
3:15
so uh I first want to talk about the uh
3:19
brilliant but lazy or
3:23
overworked reviewer uh peer-reviewed
3:26
Publications are the coin of the realm
3:27
in Academia you can't get promoted
3:30
without them you can't get a job without
3:32
them and for many fields in computer
3:35
science especially artificial
3:38
intelligence things like natural
3:39
language processing there are too few
3:42
reviewers to do a good job of reviewing
3:45
all of these papers and many people are
3:49
overwhelmed by the amount of reviewing
3:51

they have to do there are a lot of
3:52
people trying to enter the field from
3:54
industry graduate programs are expanding
3:57
and all of those students are trying to
3:59
produce res Arch and those papers have
4:01
to get
4:02
reviewed and as a result the too few
4:05
reviewers review too many papers and a
4:10
cynical way of looking at this is that
4:12
these reviewers have to find a reason to
4:15
reject a paper as quickly as possible
4:18
and so you defensively as a writer need
4:22
to make it hard for them to reject your
4:24
paper on some trivial or unimportant
4:27
basis you need to make them do their
4:31
jobs and actually engage with your paper
4:33
on a scientific level so you need to
4:35
write for these readers who are very
4:37
smart and have been in the field a long
4:39
time and can look for Flaws in your
4:42
favor and find them very efficiently so
4:45
you need to make it easy for them to
4:47
skim through your paper realize hey oh
4:49
maybe this isn't uh totally worthless
4:52
maybe we should actually accept this
4:54
paper but then they have to actually
4:56
read the paper so you need to get them
4:58
to that point these viewers are unlikely
5:01
to read your paper from start to finish
5:03
uh again they're both lazy and smart
5:06
enough to know the things that you're
5:08
going to say in like the introduction
5:09
and the conclusion so why should they

5:11
read it so you need to organize your
5:13
paper to make it easy for them to
5:16
randomly access any part of your paper
5:18
and understand what's going on so they
5:22
should be able to skim through the
5:23
Section titles of your paper and get a
5:25
sense of what's going on so don't just
5:26
call your sections introduction
5:30
methods results
5:33
conclusion that doesn't tell them
5:35
anything that could be in any paper you
5:37
need to give expressive titles
5:41
regularization improves efficiency and
5:44
you you can preface that with something
5:46
like results colon regularization
5:48
improves
5:50
efficiency tell them what's going on so
5:52
that they can get a sense of what your
5:54
paper is about similarly for every
5:57
figure give them a takeaway don't just
5:59
put a plot in and
6:02
say accuracy versus
6:04
time explain what's going on if there's
6:07
a takeaway from the figure that they
6:08
should get don't assume that they'll
6:10
spend 5 minutes looking at the figure
6:13
and realize what they need to take away
6:16
say something like the Baseline performs
6:18
better in the first 10 iterations
6:21
because it does a worse job of exploring
6:23
the
6:24
space and don't allow them to say in
6:27

their review the Baseline does better in
6:28
the first 10 ations so the method
6:32
proposed in this paper is worthless
6:34
similarly it needs to be clear what it
6:37
is that you're actually contributing in
6:39
this paper so if it's unclear what your
6:41
contribution is what your algorithm is
6:45
you need to in each of the figures uh
6:49
make sure that the reviewer knows what
6:51
it is that they should be comparing to
6:54
whatever else don't just give a long
6:56
list of algorithms where they might not
6:58
be able to decipher it similarly make
7:01
sure that your algorithms are named well
7:03
if you have many variants make sure that
7:06
it comes across which variants do want
7:09
and don't just give them inscrutable
7:11
acronyms that only you know the uh
7:14
meaning of another thing you need to do
7:17
to appease these very smart but time
7:20
constrained researchers is make sure
7:22
that you have adequate coverage of the
7:24
literature make sure that you connect
7:27
not just to the latest neural network
7:29
model models but also the probabilistic
7:31
models of the last decade and connected
7:34
to the fundamental questions of
7:36
computational linguistics don't just
7:39
have a myopic view of the literature
7:41
make sure that you connect it in the
7:43
ways that it needs to be connected for
7:45
example if uh your reviewer uh wrote the
7:49
simal paper introducing the task that

7:52
your fancy neural network is trying to
7:55
solve and you don't site that original
7:57
task definition
7:59
uh that's probably a bad Omen for your
8:03
paper the next audience that I want to
8:05
talk about is
8:07
the not overworked but maybe not all
8:10
that experienced researcher think about
8:13
like a second or third year grad student
8:16
who is reading your paper after it's
8:19
been published or maybe even reviewing
8:22
it that's becoming more and more common
8:24
these
8:25
days and they have a lot of time to
8:28
spend on your p they don't know the
8:30
field as well as professors who have
8:34
been here for a couple of decades so
8:38
they don't know the broad expanse of the
8:40
literature but they know the details and
8:44
the particulars of your problem very
8:47
well and you've probably been in this
8:51
kind of position you've had to implement
8:53
some existing work or try to understand
8:56
some existing work and you look looked
8:59
the paper and you got confused so you
9:02
need to put yourself in the shoes of a
9:07
struggling researcher who just wants to
9:09
understand the details of what you did
9:13
and this may be for
9:16
reproducibility they're coming along in
9:18
a couple of years and and trying to get
9:20
your test
9:22

train uh correct trying to do the same
9:25
pre-processing that you did and don't
9:27
just leave these details to source code
9:29
source code is great but not everybody
9:31
can use the same programming language
9:33
that you do and in 20 years that source
9:36
code may be completely useless or have
9:39
vanished and only the paper Remains the
9:42
paper still needs to be useful for
9:44
addressing these
9:45
details and for
9:48
reviewing most of the time your source
9:50
code is not in a pretty State at the
9:52
time you submit the paper and a lot of
9:54
the technical questions that these sorts
9:55
of reviewers will have won't get
9:57
answered by the source code they have to
9:59
look at the paper and they'll be pouring
10:01
over every sentence every paragraph so
10:05
make sure that you have the details in
10:06
there that would allow someone to
10:08
actually go in and
10:10
replicate pretty closely what you'll
10:13
actually do in your paper for this it's
10:17
important not just to write things that
10:20
sound
10:21
impressive but give details and so don't
10:25
just say that we use gradient
10:26
optimization talk about the package that
10:29
you're using if you're using p torch
10:31
just say that you're using pie torch and
10:35
uh perhaps give some details of say the
10:37
learning rate uh things like that to

10:40
make it more concrete
10:43
and omitting these details doesn't make
10:46
you appear more impressive and including
10:51
these details makes your paper a better
10:54
contribution to science one of the
10:57
things that makes it difficult for you
10:58
as a writer even though you may be a
11:01
junior grad student to write for other
11:03
Junior grad students is that in the
11:05
course of doing the research you
11:07
forgotten what you learned along the way
11:10
and when you started this research
11:11
project you were confused about some
11:13
things you figured it out it took you
11:15
some time to figure it out but you know
11:17
that cold now but you need to remember
11:20
the things that you were confused about
11:21
when you started the research project
11:23
and someone reading the paper for the
11:25
first time who hasn't been in this field
11:27
will likely have the same confusions
11:28
that you did and so you need to explain
11:31
that in the same way that would have
11:33
helped you when you were starting the
11:36
project so think about if you had a time
11:39
machine to go back and talk to the
11:43
person who started the research who
11:46
doesn't know all the things that you
11:47
know now what would you tell them to
11:50
make their life easier and to remove
11:52
those confusions as efficiently as
11:54
possible speaking of confusion I want to
11:56

talk about the last audience that you'll
11:59
likely be writing for members of the
12:02
media or the general public who may not
12:04
have a good idea of what's going on in
12:06
your field and might get the wrong idea
12:08
from reading your paper as written for
12:12
experts and there are two reasons that
12:15
it's important
12:16
to keep this audience in mind the first
12:20
reason is that you don't want your
12:23
research to be misconstrued you don't
12:26
want a reporter to look at your report
12:30
from a project or a published paper and
12:35
write some misleading story about it
12:37
that
12:38
either uh makes it seem like what you
12:40
were doing was stupid or trivial or on
12:44
the other hand make it seem like
12:46
computers are going to destroy Humanity
12:49
when in all
12:51
likelihood you probably have some real
12:54
scientific
12:55
contribution but it's not going to in
12:58
all probability change the world anytime
13:01
soon so to make that clear don't
13:05
overclaim make sure that your claims are
13:07
grounded in reality and if you connect
13:09
it to a real
13:11
application explain the actual ways that
13:14
it would change that application so if
13:17
you're talking about dialogue processing
13:20
don't say that it will allow Alexa to
13:23
understand uh your needs and your

13:25
motivations say that in situations where
13:29
context matters to decide the user's
13:32
intent it will improve a personal
13:36
assistance ability to detect the intent
13:39
be precise don't use words that are in
13:43
the general lexicon without explaining
13:46
the difference uh between the general
13:48
lexicon and how you're using them
13:51
specifically so for example if you talk
13:54
about regret Define regret in the
13:56
context of reinforcement learning and
13:59
online learning talk about is the
14:02
difference between the optimal reward
14:06
and what your algorithm
14:09
learns and don't let reporters uh assume
14:14
that this is now a computer feeling
14:17
regret that they missed out on uh some
14:22
emotional event the other reason that
14:25
you should try to keep this audience in
14:28
mind when you're writing your papers is
14:30
that sometimes you won't have expert
14:33
reviewers you'll have reviewers who come
14:35
from a neighboring field
14:38
or uh just don't have the the training
14:41
that they need to review the paper this
14:42
is an ideal situation but it happens and
14:46
if you can write a paper that is
14:47
engaging to someone in the
14:50
general uh populace not a scientist in
14:54
your exact area you can make it engaging
14:57
you can explain what's going on and you
14:59
can make it seem like you actually
15:00

achieve something
15:02
worthwhile even if they're not in your
15:04
area and they can get excited about it
15:07
that makes for a much stronger paper and
15:10
if you can
15:11
simultaneously make uh a professor who's
15:14
been in the field for 30 years happy you
15:17
can give the details uh to lessen the
15:20
confusion of
15:23
a grad student just entering the
15:26
field and you can
15:29
uh make a reporter excited if you can do
15:32
all those three things and appeal to
15:34
those three audiences you have a really
15:36
strong paper but how do you know if
15:38
you've actually done this and and that's
15:39
the hardest thing and and you can
15:42
internally simulate these audiences to
15:44
some degree but the only true way to
15:47
know if you've done this is to get other
15:48
people to read it and as a result this
15:51
means that you have to finish your paper
15:53
uh ahead of time and and not just write
15:55
until midnight of the deadline so one
15:58
way way that that we do this in our
16:01
research group is we have paper clinics
16:03
and this helps simulate uh overworked
16:06
knowledgeable reviewers we we have our
16:08
lab with grad students and professors we
16:10
pass the papers around and people only
16:13
have 20 minutes to read the paper and
16:14
you see what they took away and often
16:16
Times They Come Away with the wrong

16:18
impression and these are the kinds of
16:19
people who will be reviewing your paper
16:21
and if you can do this process uh with
16:24
your co-workers or your colleagues uh
16:27
that can simulate what a reviewer would
16:29
actually do and you can also turn to
16:31
your friends and family so people in
16:33
different fields of computer science
16:35
have them read what you're writing and
16:36
see if they understand what's going on
16:39
and if they do that's a really good sign
16:42
that you're writing well it may not mean
16:44
that you have the greatest research but
16:46
if you can explain it well that counts
16:48
for a lot and a well-written paper is
16:52
more likely to get good reviews which
16:54
we'll talk about next

How to Review a Paper

we're going to continue our discussion
0:01
of how to write papers for machine
0:04
learning and natural language processing
0:06
but today we're not going to talk about
0:08
writing per se we're going to talk about
0:10
what happens after you finish writing
0:11
what happens when you submit your paper
0:14
to a conference or to a journal and
0:18
reviewers in your field decide whether
0:21
it should be published or not and even
0:23
though it's not writing per se keeping
0:28
these factors in mind as you're writing
0:30
can lead to a better paper with a higher
0:32

probability of acceptance at the kind of
0:34
venues and what your paper to be
0:35
accepted after and oftentimes this is
0:40
viewed as an adversarial process that
0:43
the reviewing process is trying to find
0:46
ways for your paper to be rejected and
0:48
it's a painful soul-crushing experience
0:51
and it can be that I won't lie about
0:55
that but it's really exposing your paper
0:59
to the first readers these are people
1:01
chosen by experts in your field who can
1:03
decide whether your paper is good or not
1:05
and because you want to write for your
1:10
readers it's important to write for
1:13
these very first readers of your paper
1:15
and to make sure that they can
1:17
understand what your papers about so
1:19
that they can write a clear and thorough
1:21
review so understanding this process
1:23
will help you do a better job of writing
1:26
a good paper and getting through the
1:27
review process and for yourself learning
1:32
how to write a good paper and be a good
1:34
reviewer is an important part of
1:35
academia like much of academia the
1:38
reviewing process is hierarchical and so
1:40
we're going to talk a little bit about
1:42
the hierarchy of how papers get reviewed
1:44
for a conference it's a little different
1:46
for journals but many of the same
1:48
concepts apply so for journals you have
1:51
editors action editors and individual
1:54
reviewers all of the same concepts that

1:57
you see in conferences it's about
1:58
because we're in computer science we're
2:00
going to talk more about conferences at
2:02
a conference you typically have at the
2:05
very top of a set of program chairs and
2:09
so these are the people who are in
2:11
charge of the
2:12
haier program the list of accepted
2:14
papers that this conference will have
2:16
and then they will divide the conference
2:20
into areas and so these areas are areas
2:23
of expertise something like language
2:28
models or topic models or parsing areas
2:31
of a conference where experts can judge
2:34
whether this set of papers is good or
2:37
not and then each of those areas will
2:40
have a bunch of reviewers and those
2:43
reviewers will decide forever giving
2:45
paper whether it should be accepted or
2:48
not
2:49
it will write a review that's why
2:50
they're called reviewers at very large
2:52
conferences like ICML you'll have
2:55
another layer in between so instead of
2:57
just having area chairs you have seen
2:59
your area chairs area chairs below them
3:02
and those area chairs interacting with
3:04
reviewers so this has another layer to
3:06
the hierarchy and in many cases the
3:10
hierarchy also expands below so that
3:12
instead of just having individual
3:15
reviewers looking at a paper those
3:17

reviewers may call it assistant
3:18
reviewers and for example I may review
3:21
at a conference and I may see a paper
3:23
very much related to something that my
3:25
student is working on I will give that
3:27
student the paper and ask them to be an
3:29
assistant reviewer in both of our names
3:30
will go on the resulting review so one
3:34
question you might ask is how do these
3:37
people get selected the area chairs are
3:39
selected by pan client the program
3:41
chairs and those area chairs go out and
3:44
look for people who have published in
3:45
this area before people who viewed in
3:47
this area before and they'll also look
3:49
for senior students who have published a
3:52
couple of papers and are ready to start
3:53
reviewing on their own this is not an
3:56
obligation you will be asked do you want
3:58
to review for this conference and the
4:00
entire scientific process depends on
4:03
review we're saying yes and there are a
4:05
lot of commitments out there but
4:06
reviewing is an important part of being
4:08
a supportive and contributing member of
4:11
the scientific community so you have to
4:13
say yes but only say yes if you can
4:16
actually do the job so the worst part of
4:20
being an area chair is having
4:22
of reviews come in late or come in with
4:26
only a couple of words and these are
4:29
really disruptive to the process I'm
4:31
sure that you as authors would hate it

4:34
if you were expecting to get three
4:36
reviews and you got one good review and
4:40
then one one liar of you and then a
4:41
missing third review and that means that
4:45
your paper wasn't given the fair
4:47
treatment that it deserved so doing to
4:50
others that do would happen to end then
4:52
if you signed up to be a reviewer glue
4:54
good job and do the sorts of things that
4:56
we'll talk about today what important
4:58
part of being selected as a reviewer is
5:00
having a webpage so it's good in general
5:03
and academia to have a web page that
5:05
lists the projects that you're working
5:06
on in your publications but specifically
5:09
for finding reviewers having a web page
5:11
will help reviewers know if you were
5:13
ready to review or not and if you want
5:17
to be part of your reviewing process
5:19
have a web page that shows that you have
5:21
the skills that it takes to do a good
5:23
job reviewing when we talk about writing
5:25
we talked about the different kinds of
5:27
audience and setting right for and this
5:29
is also true when you write a review
5:32
when you write a review you're rhyming
5:34
not just for the authors but also for
5:38
the area chair and other reviewers and
5:40
so think about the things that you have
5:42
to do in this process to make sure that
5:44
your review is well-written for each of
5:46
these audiences so first be nice to the
5:50

authors and we talked about how
5:53
reviewing can sometimes be a
5:55
soul-crushing process when you have
5:58
submitted your life's work to a
6:00
conference or a journal and you get back
6:02
a scathing review so don't be that
6:05
person write a nice review and one very
6:08
simple way of doing that is talk about
6:09
the paper don't talk about the author
6:12
I saw one particularly bad review that
6:15
said something like the authors have
6:17
wasted our time by submitting a
6:19
half-baked idea and these are likely
6:21
junior students who are having their
6:23
first brush with academia and that's not
6:25
a good way to introduce them to the
6:28
scientific community be constructive
6:30
reviews are a chance for papers to
6:33
become better and so even if the paper
6:35
is certain to be rejected
6:38
tell the author's what they can do to
6:40
make the paper better so don't just say
6:43
the evaluation sucks say the evaluation
6:47
would be stronger if you did experiments
6:49
a B and C and then they can do those
6:50
experiments and have a better paper if
6:53
you just say the experiments suck you'll
6:56
get the same paper back that you saw
6:58
last time maybe with ten times as many
7:01
bad experiments so it takes trouble for
7:05
both you as a reviewer and trouble for
7:08
the authors you should also be concrete
7:11
and so if you are going to give a

7:14
suggestion or give it criticism make
7:16
sure that it's backed up like that so
7:19
don't just say this has been done before
7:21
it's illicitly saving the paper that did
7:24
it if you're saying you should have used
7:27
a data set that has properties a B and C
7:29
make sure that that data set actually
7:31
exists and see what that data set is
7:33
maybe you have information that the
7:35
authors don't have or maybe you're
7:36
remembering it wrong if you have a
7:38
criticism in concrete in your criticism
7:41
so that the author can either learn from
7:44
it or if you're wrong and reviewers are
7:46
often wrong they can refine it but if
7:49
you're vague it doesn't really help
7:51
anybody so your writing to the authors
7:53
to help them improve their paper but the
7:56
point of reviewing is to decide whether
7:57
this paper is getting accepted to the
7:59
conference or not and the person making
8:02
that decision is the area chair so you
8:04
also need to write for the area chair
8:06
and it's important to keep in mind that
8:08
the area chair is looking at lots and
8:11
lots of papers dozens maybe even a
8:13
hundred papers and they need to decide
8:15
which of these papers get in and which
8:17
of these papers are going to get
8:18
rejected so make the internet Jerry's
8:21
life easier and your reviews will be
8:25
treated with greater respect and you
8:28

will gain a better reputation in the
8:30
community so first off be self-contained
8:33
because the area chair is reading lots
8:36
and lots of reviews things get confused
8:39
and they don't always remember every
8:41
detail of the paper so make it so
8:43
that they can read your review and it
8:45
stands on its own don't refer to other
8:47
reviewers without fully explaining
8:49
what's going on don't refer to section
8:52
numbers in the paper become creeks and
8:55
specific enough that the area to you can
8:57
read your review and understand what's
8:59
going on don't make it so that they have
9:01
to open up two tabs with your review and
9:04
the paper side by side to understand
9:06
what you're saying and along with us be
9:10
detailed so the area chair is an expert
9:13
in the field and if you tell them what's
9:15
going on in the paper they can have a
9:18
much better idea of whether this paper
9:20
should be accepted or not so don't just
9:23
say that the equations are justified
9:25
talk about the specific problem and and
9:28
these people are experts they are smart
9:30
they'll be able to understand what
9:31
you're saying and they'll be able to use
9:33
that information to make an informed
9:34
decision but if you just say the
9:36
equations are justified they need to
9:38
spend more time to go into the paper
9:40
read through the paper maybe they can't
9:43
see the problem that you're seeing they

9:45
don't have enough time to do it
9:46
use your time and your energy that you
9:50
put into reading the paper so that the
9:52
area chairman can make an informed
9:53
decision and if there's a critical
9:55
problem with the paper that can find it
9:57
immediately and judge how big of an
9:59
issue it actually is and be thorough so
10:03
because the area chair has hundreds or
10:07
at least dozens of papers to look
10:09
through they can't find all of the
10:10
problems that you can find so if there
10:13
is an issue it's your job to find it
10:15
it's your job to raise it so look to see
10:19
if there is work that hasn't been cited
10:21
look to see if there are issues in the
10:24
derivation or in the proofs and if so
10:27
raise those issues and explore all the
10:30
places where there could be a problem in
10:32
the paper even if you don't find a
10:34
problem that's important information to
10:35
say I thoroughly check the proofs and
10:39
everything checks out as far as I can
10:42
tell that's very useful and many times
10:45
people will only mention where there's a
10:48
problem but it's very useful for the
10:50
area chair to know that someone that
10:52
they respect and trust has checked in
10:55
part of the paper and found it to be
10:56
just fine and so that's important
10:59
information for the area chairs half as
11:01
well at many conferences another
11:03

important audience for your reviews are
11:05
the other reviewers if you get to see
11:08
the other reviews engage with them and
11:10
the area chair needs to take all the
11:12
reviews for a paper and make a final
11:14
decision if one reviewer says that this
11:19
model has problems X Y & Z and you
11:21
disagree with that you need to say in
11:23
your review that those aren't problems
11:25
that those are good things and this can
11:27
happen through the discussion but it
11:29
needs to get formalized inside the
11:31
review oftentimes after the reviews go
11:34
out there will be some email discussion
11:36
and you'll talk about the paper and you
11:38
may reach some decisions but that needs
11:41
to get memorialized somehow sometimes
11:43
the area chair will write a measure of
11:46
you explaining the details but it's your
11:49
responsibility to express your thoughts
11:51
and your reviews and so you have in many
11:54
cases the option of editing your reviews
11:56
so edit your reviews to reflect the
11:58
discussion that you had with the other
11:59
reviewers and this is another
12:02
opportunity for you to be constructive
12:04
and informative and detailed in a way
12:07
that can helping authors
12:08
and this helps the area to hear make a
12:11
better decision if they can see all of
12:13
the discussion reflected in the reviews
12:15
in a nice formal way that doesn't always
12:18
come across and reading through the

12:19
emails and implicit in all of this is
12:22
that you have to actually engage in the
12:24
discussion don't consider your job done
12:26
when you submit the reviews and then
12:28
forget about them and again doing to
12:31
others as you would have them do unto
12:32
you once you submit a review if there is
12:37
feedback from the authors or from other
12:40
reviewers or the area chair that should
12:42
be reflected in your review and that
12:44
will go to the authors and they'll be
12:46
able to improve the paper with that
12:47
feedback so even though you as a
12:50
reviewer will make a recommendation
12:51
should this paper be accepted or should
12:54
it be rejected that's not the end of the
12:56
story your recommendations flow up and
12:59
so the area chair will read through all
13:02
the reviews and then they'll make a
13:03
recommendation
13:05
and then the area chair will usually get
13:07
a sorted list to the program chair and
13:09
then the program chair will decide these
13:12
are the papers from each area that will
13:14
accept and different areas to have
13:16
different levels of strengths so from
13:18
the ranked list of each area chair
13:20
provides the program chair will decide
13:22
here the accept line is and where the
13:24
reject line is so think about how you
13:28
can help the area chair do a good job of
13:31
recommending a paper for acceptance or
13:34

rejection so the best thing possible is
13:37
for your review to be both decisive and
13:41
detailed so say I believe that this
13:45
paper should get accepted and here's why
13:47
or I think this paper to get rejected
13:49
and here's why second vest is for you to
13:54
be ambivalent in your decision a
13:57
detailed interview and who may not have
14:00
the experience to say whether it's more
14:04
important to have a good problem set up
14:06
and a quality evaluation or vice versa
14:09
and you are not comfortable making that
14:12
position but if you lay out the pros and
14:16
cons of the paper that's something that
14:18
an area chair can work with to come to a
14:19
decision and so they have the
14:21
information that they need and they can
14:24
be the decisive one but it's bad if
14:27
you're reviewed as sparse on details and
14:30
so even if you're decisive and say this
14:33
is a great paper and that's what you say
14:35
in your review the area chair can't do
14:38
much with that because they'll need to
14:39
make an argument and they'll need to
14:41
reconcile your review with the other
14:42
reviewers and so if there were other
14:45
negative reviews that were far more
14:47
detailed where review isn't going to
14:49
count for much and the authors aren't
14:52
going to get much out of that and
14:53
similarly if you're ambivalent and you
14:56
have a very sparse review your review
14:58
will get ignored for good reason you

15:00
don't say what you think and you don't
15:02
explain why you thought that so if you
15:06
want to be a value of reviewer make sure
15:08
that your reviews are detailed so that
15:10
the area chair can make a decision even
15:12
if you can't make a decision
15:13
work for you to give the tools for the
15:16
area chair to make that decision so even
15:19
if you never review it's useful to think
15:20
about what happens when a paper gets
15:23
reviewed and this will help you as
15:26
you're writing so you can think about if
15:29
I were reviewing this paper
15:31
why would I reject this paper we talked
15:33
about those before in modeling the
15:35
reader who is lazy but experienced and
15:38
as you think about the process for how
15:41
papers get accepted into a conference
15:43
this will help you write a stronger
15:44
keeper that more immune to the kind of
15:47
things that can go wrong during the
15:48
reviewing process and think about these
15:52
specific people who might be reading
15:53
this paper if you're building on an
15:55
existing paper the area chair may well
15:59
assign the people who worked on that
16:01
paper to review it so talk about those
16:05
papers and collegial scientific ways and
16:10
think about the other people who work in
16:13
this area have you cited them have you
16:15
explained how your work is different if
16:17
there are related papers who do similar
16:20

things those are likely to be your
16:23
reviewers and they understand their
16:26
system so if you cite their paper and
16:28
explain why your system is different
16:30
that will help them do a better job of
16:32
reviewing the paper
16:33
think about how you would explain it to
16:35
them if you saw them at the later
16:37
conference and that's the kind of thing
16:39
that should go into the paper the same
16:41
way that you would explain it to them in
16:43
person is how you should explain it to
16:45
them in your paper and because your
16:48
paper will be reviewed in a pool of
16:50
other papers think about the other
16:52
papers that are likely to be submitted
16:55
to this conference what about your paper
16:58
is special or unique and if you're
17:00
submitting yet another data set that is
17:03
question-answering
17:04
what about your paper this year is going
17:07
to be unique and set it apart from
17:09
everything else the bottom line is that
17:12
the scientific community is a community
17:14
and it's full of individual people who
17:17
are reviewers writers area chairs
17:20
program chairs they're all human they
17:22
all make mistakes but science is a
17:26
process of communication and everyone
17:29
trying to get to a better degree of
17:31
understanding and the more effective you
17:33
are at writing and reviewing the more
17:36
successful you'll be in science and the

17:39
faster that we as a community understand
17:42
the difficult problems and challenges
17:44
that our patients

The Grammar of Graphics

I'm a computational linguist.

0:04
We make a lot of graphics to explain our research. And some of our research is on grammars!
0:12
Today we're going to talk about the "grammar of graphics", my preferred way of creating graphics.
0:19
Because many of you know the linguistic definition of a grammar, you know that
0:22
a grammar is a recursive definition of rules that defines a means of communication
0:27
where non-terminals transform into non-terminals and terminals. Now, technically, the grammar of
0:35
graphics isn't a grammar in this sense: it's not recursive, doesn't have non-terminals, and so on.
0:42
But it is about communication and transformations.
0:44
The Grammar of Graphics was developed by Lee Wilkinson, who died recently in December 2021.
0:50
In a tribute to him, Jessica Hullman summed up his contributions to the field.
0:55
And her tribute captures two of the ways that the grammar of graphics line up with the ways that
1:04
linguists think about grammars: equivalence through transformations and grammaticality.
1:09
The first relationship between how we think of grammars and the grammar of graphics I'd
1:13
like to highlight is that grammars allow for transformations. Say you have the sentence
1:18
"the ball hit Mary" and you'd like to turn that into a question that asks
1:22
"whom did the ball hit". In some sense this is the same underlying sentence, but it's
1:29
ordered differently. In many languages, you have wh-movement to put questions words first,
1:35
and the verb has changed. But the underlying idea is the same.
1:42
The same holds true for graphs. Even though the pixels of these two charts are very different,
1:48
the underlying data is the same. The elegance of the grammar of graphics
1:55
is that this changes two aspects of the graph: the aesthetics of the colors of the elements
2:08
and the coordinate system that the plot is using. But the underlying data are the same.
2:13
And for someone making a plot, this is important. You might need to explore
2:17
lots of different transformations and presentations to understand
2:22
what the heck is going on in your data. The grammar of graphics makes this easy.
2:30
The flip side of what a grammar allows is what it doesn't allow you to do. In language, we call this
2:36
a sentence that is not licensed or ungrammatical. If you can't derive it in the grammar, it's not a
2:46

valid sentence. So you can't say things like "hit ball Mary the".

2:50

Similarly, one of Lee Wilkinson's insights is that a lot of mistakes in visualization data were

2:57

a deviation from the grammar of graphics. Doing a transformation at the wrong time, or applying

3:05

statistics inconsistently. You go in this order or you don't go at all.

3:11

For users who are new to the grammar of graphics this can be frustrating. You

3:16

want to do something ... you need to figure out what stage of the grammar it would lie at, what

3:22

the commands are, and how it interacts with the other transformations. Why is it so complicated?

3:30

Part of the complication keeps you on the straight and narrow... other plotting packages give you

3:37

access to raw pixels or vectors. It's easy to do anything! But it's also easy to make bad plots.

3:48

In the next video, I'll talk about the sequence of creating graphics.

3:51

As I go through the steps, keep this figure in mind as we build up visualizations

3:58

in implementations of the grammar of graphics like ggplot2 or plotnine.

4:04

So this is why I recommend learning the grammar of graphics.

4:07

Just like natural language, it's about communication. And good communication

4:11

requires structure and rules... if you play by the rules, you'll get your point across.

4:16

And using some of the packages we'll talk about, you'll look good while doing it.

Introducing Plotline

thus far we've talked about how to read

0:02

in data and later we're going to talk

0:05

about how to build models to make

0:06

predictions on that data but something

0:09

really important that comes before you

0:11

make predictions and after you make

0:13

predictions is understanding what's

0:14

going on inside your data and to do that

0:16

you need to be able to look at your data

0:18

and to understand what's going on and

0:20

you often can't do that by looking at

0:22

the raw data themselves you need to

0:23

visualize them in some way now you can

0:26

take an entire course on data

0:28

visualization and I encourage you to do

0:29
so but we're just going to cover the
0:32
basics in this course so that you can
0:34
get a sense of what's going on your data
0:36
and perhaps diagnose problems the reason
0:38
that we're talking about it in a data
0:40
science class is that you simply cannot
0:42
build a good model on noisy data and
0:45
once your day to get large enough it's
0:47
impossible to go through a file and look
0:50
at the data and understand what's going
0:52
on and where there are problems and
0:54
while we may be able to do that for
0:56
hundreds of observations once you get
0:58
into thousands it's simply impossible to
1:00
find patterns and discover issues on
1:03
your own without some sort of assistance
1:05
and that's where data visualization can
1:07
come in you can understand what's going
1:08
on inside your data so that as you build
1:11
a model later you can detect problems
1:14
and you can see how well your model
1:16
lines up with the reality of the
1:17
underlying data we're going to be

Grammar of Graphics

1:19
talking about a particular formulism for
1:22
doing visualization called the grammar
1:24
of graphics and the reason I like the
1:27
grammar of graphics is that it doesn't
1:28
focus on manipulating individual pixels
1:31
in an image even though it gives you a
1:32
level of generalization it's still
1:34
relatively powerful and lets you create
1:36

really compelling graphics this
1:39
framework is most commonly used in an AR
1:42
package called ggplot2 and we're going
1:46
to be using some of the examples from
1:48
Hadley wickham's introduction to that
1:52
book which is also part of the reading
1:53
for this week but we'll be using a
1:56
Python package called plot 9 that has
1:59
many of these same features it's not
2:01
quite as robust or as well documented as
2:03
do
2:03
- but I'm not going to ask you to learn
2:06
a different programming language our if
2:08
you're already very comfortable building
2:10
models and cleaning data in Python there
2:13
are many reasons I like this framework
2:14
for building graphics one reason that I
2:17
like it is that you can very easily plug
2:21
different visualizations together or
2:24
different components of visualizations
2:26
together and this allows you to create
2:29
really advanced graphics from very
2:30
simple building pieces and it also has
2:33
reasonable default and so if you're just
2:35
getting started
2:36
say for example and of course like this
2:38
if you do something it won't be too
2:41
crazy and you can use that to slowly
2:44
refine your graphics over time to create
2:46
something really interesting or to
2:48
understand what's going on inside your
2:50
data so let's start with a really simple
2:51
data set we have four observations two

2:56
continuous coordinates that we'll call X
2:59
and Y and one discrete coordinate that
3:04
will call the shape and what we want to
3:07
be able to do is to be able to figure
3:09
out the two different types of data that
3:11
we have in this data set the A's and the
3:14
B's and how we can tell a story with
3:17
those data and in particular how do we
3:21
contrast those two groups with each
3:23
other so one thing that the grammar of
3:26
graphics allows us to do is to transform
3:29
components of our data into things that
3:32
make sense for a visualization so for
3:34
example we may want to turn the A's and
3:36
B's into different shapes and we'll
3:38
display them somehow in a graphic and so

Geometry

3:42
in the language that ggplot2 and plot 9
3:45
use these are called geometries so the
3:49
geometries are the shapes of the images
3:52
that will be displayed on your screen
3:54
you can also have things like coordinate
3:57
systems that tell the values of the
4:01
elements that you're attaching to a plot
4:03
and then you can also have annotations
4:06
like say the axes on a plot so let's see
4:09
a really example of doing that within
4:12
plot 9 and so at this point we have a
4:16
pen
4:16
dataset called demo and so demo is our
4:22
data set and it's just a panda's data
4:24
frame with the same columns that I
4:29

showed you before XY and shape so what
4:33
we're doing here is we're creating a
4:34
grammar of graphics object and so we do
4:37
that by calling this ggplot function and
4:41
then we tell it what data we're going to
4:44
look at and here we're saying what
4:46
aesthetic we're going to be using and
4:48
here we're going to say plot these
4:50
points and choose a color based on the
4:54
shape and then use Y as the y coordinate
5:02
and X as the x coordinate and so these
5:06
get plotted here and so the a shapes are
5:09
down here the B shapes are up here and
5:13
we've also told it that we want these to
5:16
be points we want to plot these as
5:18
points and say a scatter plot then we
5:22
tell Python let's save this to a file
5:24
and then it saves it to a PDF that you
5:26
can then insert into a document that
5:28
you're writing up or you can send it in
5:30
an email or whatever else you want to do
5:32
with it the nice thing about the grammar
5:34
of graphics is that you can very easily
5:36
swap out different components to create
5:38
very different visualizations so if you
5:41
want to use something other than points
5:43
to plot your data you could do something
5:45
like plotting a line and here we have
5:52
swapped out geometry point G employed
5:55
for G online so this is the only thing
5:59
that's changed here and we have a very
6:01
different graph as a result there are
6:03
also many other geometries that we could

6:06
use I'm not going to show you all of
6:07
them but another common one is using a
6:09
bar geometry and so here you say use
6:12
geometry bar G on bar but you also need
6:16
to tell it the how tall to make that and
6:20
so here you're saying just use the value
6:23
in the y position for how tall that bar
6:26
is going to be and that's basically
6:29
saying use the ident
6:30
the function to figure out how tall the
6:32
bar is going to be the nice thing about

Aesthetics

6:35
these frameworks is that when you choose
6:38
to apply some aesthetic to a point it
6:42
makes reasonable selections for you so
6:45
for example if you tell it the size of a
6:48
point and that's associated with
6:49
continuous data it will size a ball or a
6:53
square whatever shape is associated with
6:55
it to a size proportional to whatever
6:59
value is associated with that column so
7:01
we could have another column in our data
7:04
set that represents population or
7:06
frequency or something like that and in
7:09
addition to the XY coordinates we could
7:11
say how big of a dot is it corresponds
7:14
to this additional parameter if we
7:16
assign a color to a continuous parameter
7:19
then it goes on a spectrum
7:21
say for example from red to blue and the
7:25
larger the value is the more blue that
7:27
against the smaller the value it is the
7:29

more red it is and so here again we're
7:31
associating multiple variables to a
7:36
single point that's being displayed and
7:38
we can associate that with either size
7:40
or color in addition to the standard XY
7:42
coordinates we'll see an example of this
7:44
in just a second you can also do similar
7:46
things for discrete data so for example
7:49
you can have different shapes associated
7:52
with different categories of data or
7:55
different colors and here if you
7:57
associate a color with discrete data it
7:59
does something very sensible instead of
8:02
using a spectrum it now uses as
8:05
different colors as possible to show you
8:08
the distinctions between these different
8:11
categories so you may have a plot that

Scale

8:13
looks like what's going on on the left
8:16
here all your data is bunched up and you
8:20
may want to transform one or both of
8:22
your axes to use a different scale so
8:25
instead of a linear scale you may want
8:27
to use a logarithmic scale
8:28
this can help expose for example
8:31
exponential relationships and your data
8:33
into something a little bit more easy to
8:36
digest and see with our eye so for
8:39
example if you're looking at any sort
8:40
money
8:42
new data often it's very sensible to
8:44
apply some sort of logarithmic scale
8:46
because as prices get really really big

8:49
they can make it hard to see patterns
8:51
throughout the entire range of your data

faceted graphs

8:54
let's go back to our very toy data set
8:57
that we had before we're trying to
8:59
compare these two things the the circles
9:01
and the squares another thing that you
9:03
can do to help show the differences is
9:06
to use what's called a faceted graph and
9:09
so let's say that you have two states
9:14
Maryland and Virginia and want to
9:15
compare them you could show the same
9:17
data for both states side-by-side next
9:20
to each other and the nice thing about
9:21
this is that now these graphs have
9:23
consistent axes you can tell what's
9:26
going on and you can very easily compare
9:29
left and right to see how these two
9:32
different data sets compare with each
9:34
other sometimes it makes sense to try to
9:36
do that in one graph using things like
9:39
color or shape to differentiate the two
9:43
groups other times putting it
9:44
side-by-side tells a clearer picture
9:46
many times you have so many variables is
9:49
just simply impossible to put it all
9:51
into a single graph as I said before
9:53
it's possible and probably even a good
9:55
idea to take a whole course in data
9:57
visualization we don't have time to go
9:58
over all of that and because we're
10:01
working in Python I want to give you
10:03

enough of a brief introduction so that
10:05
you can play around and discover trends
10:08
and data visually and this will be
10:11
something that's really important to do
10:13
before you build a model and for things
10:15
like feature engineering that we'll talk
10:16
about later we'll use plot 9 in this
10:19
course because it's a Python package if
10:21
you're using R for any reason please
10:24
take a look at ggplot2 it's a very
10:27
full-featured and many of the things
10:29
that you can do in ggplot2 you can also
10:31
do in plot 9 with very minimal changes
10:34
to the syntax next we'll be talking
10:37
about examples of what you can do to
10:40
build nice visualizations in plot 9 and
10:42
also what sorts of things you want to
10:44
stay away from in creating plots that
10:47
are less useful in conveying
10:50
seeing information

Visualization Example

now that we've had a really brief
0:01
introduction to plot 9 and visualization
0:04
let's walk through an example of
0:06
creating a visualization and the
Napoleons Grand Army
0:10
visualization that we'll focus on is a
0:13
very famous one so this is a
0:16
visualization I created before computers
0:20
and this was created by someone named
0:23
Charles Joseph Menard and Edward Tufte
0:27
a-- who is kind of the guru of data

0:30
visualization and telling stories with
0:32
data praise this as one of the best
0:36
visualizations of data ever created and
0:39
this was created before computers before
0:43
Python before plot 9 this was just done
0:46
by thinking very thoroughly about a
0:49
problem and then sitting down with a
0:51
blank sheet of paper a ruler and a lot
0:56
of insight into the story that he was
0:58
trying to tell and the story that he was
1:00
trying to tell was about what happened
1:03
to Napoleon's Grand Army when it invaded
1:07
Russia in the early 19th century and so
1:10
Napoleon assembled a huge army called
1:14
the grand army and sent it in to Russia
1:17
and there were a number of battles
1:21
Napoleon did relatively well at first
1:24
but as time went on his men were
1:27
whittled down slowly slowly slowly as
1:30
they got deeper and deeper into Russia
1:32
and eventually the Russian winter and
1:36
the courageous fighting of the Russian
1:39
troops forced the Grand Army to retreat
1:43
this visualization tells that story as
1:46
you go from left to right from west to
1:51
east you can see visually the columns of
1:55
men slowly shrinking down to a very
1:58
narrow point and then you can see the
2:01
armies turning back around and heading
2:04
in the opposite direction and the width
2:07
of the line represents how large
2:09
Napoleon's army ones and you can see
2:13

as the army returns back it's splitting
2:15
up and diverging to different
2:19
destinations and so you have both time
2:23
and space being represented by the
2:26
x-axis and you have the width of the
2:31
line representing the number of men and
2:34
one thing that we won't do in our
2:36
visualization is at the bottom you also
2:37
have the temperatures and you can see
2:39
the temperatures getting lower and lower
2:40
and lower and you can see the toll that
2:44
the temperature is taking on the number
2:46
of troops so here's what the data looks

CSV Data

2:49
like if we save it as say a CSV and so
2:54
we can call the columns the longitude
2:58
the latitude the number of troops the
3:01
direction that they're going on are
3:04
advancing or retreating
3:05
and the division and so recall at the
3:08
end as the retreating Napoleon's army
3:11
breaks off into smaller groups this
3:13
allows us to keep track of that and so
3:15
here's our data formatted in a way that
3:18
a computer can make sense of and so
3:20
recall that the latitude and longitude
3:21
are ways of dividing up our world into
3:26
coordinates and so if you use the GPS
3:29
devices telling you your latitude and
3:31
longitude and these are basically the
3:33
GPS coordinates of Napoleon's army but
3:35
that isn't the only data that's being
3:37
reflected here there are also the names

3:39
of the cities in those cities represent
3:41
battles or destinations for Napoleon's
3:43
army and those two have latitude and
3:46
longitude positions and you have the
3:48
name of the city you can also add an
3:50
additional information here like the
3:52
populations of the cities and then you
3:54
could use say like the size aesthetic to
3:56
show the size of those population

Plotting Data

3:59
centers so here's a first pass at
4:01
plotting these data and so we've loaded
4:04
in this data frame through pandas we
4:07
have also imported in all of our
4:09
commands from plot 9 that command looks
4:12
something like and we have a panda's
4:16
data set where we've loaded in the CSV
4:18
file that I just showed you for the
4:20
troop sizes and what we're going to do
4:23
is we're going to
4:26
plot these points using the longitude
4:31
and latitude for the x and y coordinates
4:34
and then we're going to use the path
4:37
geometry and that's done just by this
4:40
plus sign and then saying let's create a
4:42
path on our plot where the size is the
4:47
number of troops and so the width of the
4:48
line represents how many troops there
4:50
are and then the color represents the
4:53
direction so cyan for retreating red for
4:57
advancing and so advancing in this
5:02
direction retreating in this direction
5:03

and then we also have additional
5:09
divisions and so this is going to plot
5:12
these separately so after we've done
5:14
this we have something that looks a lot
5:16
like that visualization we have the
5:18
width of the lines representing the size
5:21
of the troops we have different colors
5:23
for advancing and retreating and we have
5:27
the different branches of the army
5:30
representatives different paths but what

Text Geometry

5:32
we're lacking is important context and
5:35
so this XY coordinate space doesn't mean
5:38
much without the cities involved and so
5:40
we can add in the cities using a
5:43
different geometry and so there's a text
5:45
geometry where you can add text
5:47
explicitly to your plots and this is
5:50
going to use the text in the city column
5:53
from our cities pandas file and this is
6:01
just saying use it in a pretty big font
6:04
and so then the city names appear on our
6:07
plot and so you have Moscow all the way
6:10
here at the end and you have Minsk here
6:16
in the middle and so this is giving you
6:18
a historical context for what's going on
6:21
if you're a student of the Napoleonic
6:23
Wars you can recognize some of these
6:25
place names and understand why there was
6:29
a big decrease in the size of the army a
6:32
particular point
6:33
sometime there are a lot of things you

Other Ideas

6:35
could do to make this map even prettier
6:37
so for example you could add the
6:39
outlines of modern-day countries you
6:43
could annotate specific battles you
6:47
could annotate dates you could put the
6:50
temperature like we saw before would say
6:52
a faceted graph all of those would be
6:55
interesting and fun to do but I'm not
6:58
going to do too much but what I am going
7:00
to do is I'm going to tweak the colors a
7:01
little bit so that we have an emphasis
7:07
on the retreating army and you have that
7:11
in like a blood-red color to sort of
7:16
make that a little more visceral so that

Conclusion

7:18
is a very brief introduction to how you
7:21
can create plots using plot 9 this is a
7:24
derivative of the original Menard plot
7:26
that tooth - like so much I used data
7:31
from Hadley Wickham who use this as an
7:34
example of ggplot2 and then I adapted it
7:36
to plot 9 so there's been a number of
7:40
translations over time and hopefully you
7:44
can play around with it yourself and
7:46
create your own version of this
7:49
visualization as you're doing this as
7:52
you're playing around with it make sure
7:54
that you do have plot 9 installed you
7:57
can use the pip utility that comes with
7:59
Python also make sure that you have the
8:01
data saved in the appropriate location
8:03
just like for the previous exercises
8:06

that we've done you need to be able to
8:08
access the data to read it in through a
8:11
panda's one issue that often crops up is
8:16
that people use the command line to
8:18
generate quick-and-dirty plots I
8:20
encourage you not to do that I encourage
8:21
you to save all the plots that you
8:23
create both the commands that you use to
8:25
create them and then have a file that
8:29
you can run automatically to regenerate
8:31
those plots and so no plot is perfect
8:34
and you will often need to go back
8:35
multiple times to correct it or refine
8:38
it and so make sure you save your work
8:40
and sometimes you'll have a really
8:42
beautiful plot that you want
8:44
have that same sort of thing for a
8:47
different dataset and so you always want
8:49
to be able to go back retrieve the
8:51
commands you use to create a plot and
8:52
then create a new plot that looks
8:54
exactly like it with a different data
8:56
set and that's the beauty of the grammar
8:58
of graphics because it abstracts out a
9:00
lot of the decisions about creating
9:03
these plots you can take the same
9:05
visualization and apply it to multiple
9:07
data sets with not a lot of effort on
9:10
your part

How not to Visualize Something

so now that we've given an example of
0:02
how to create an effective visualization

0:04
let's give some examples about how to
0:06
create an ineffective visualization and
0:09
so I've collected examples from other
0:13
people who have collected examples of
0:15
bad visualizations and we'll talk
0:17
through about what makes a bad
0:18
visualization so that you can try to
0:20
avoid that as well before we get into
0:24
specific examples let's talk about some
0:28
overarching principles so try to avoid
0:31
clutter and oftentimes when you have a
0:33
lot of data and you run the default
0:36
visualization you get a lot of stuff and
0:39
that stuff may not all be important to
0:43
the story that you want to tell and part
0:45
of telling a good story is a principle
0:48
of less is more and as you remove or
0:51
de-emphasize aspects of your data you
0:54
may be able to draw the readers eye to
0:56
what's actually important because you'll
0:58
have to display a legend and that legend
1:01
should make sense the order should
1:02
correspond to say the average value of
1:06
that category or it should be
1:09
chronological don't disorder
1:12
alphabetically which is a reasonable
1:13
default that you're plotting software
1:15
will do because it doesn't know any
1:17
better make sure that you enable the
1:19
reader to make comparisons so your
1:21
reader will want to know the difference
1:22
between a and B so be able to highlight
1:27

the different things that they should be
1:29
comparing so use different shapes use
1:31
different colors use different lines all
1:34
of these things together to make sure
1:36
that the reader can make comparisons
1:39
between various things that they may
1:42
want to compare and those things may not
1:43
correspond to what you want to compare
1:45
you may be trying to introduce some new
1:48
model and you want to show that that
1:50
model is great but maybe you're
1:52
comparing it against Sarah's model and
1:56
Sarah will want to be able to compare
1:58
her model to all the other models you
2:00
ran to see how her model did so make
2:03
sure that you enable these comparisons
2:05
and also think about the different ways
2:08
that
2:09
or visualization will be consumed and so
2:14
not everyone will view it on say a huge
2:19
screen monitor people may be viewing it
2:22
on a smartphone they may print it out on
2:24
a black-and-white printer make sure that
2:27
your visualization will work in all
2:28
those formats as best as possible so
2:30
before I talked about the quantity of
2:32
data there's also how many things you
2:35
can put inside a single plot and again
2:38
often less is more
2:40
don't try to tell all the stories that
2:41
you want to tell with your data set in a
2:43
single plot sometimes you'll want to use
2:46
different plots that tell different

2:47
stories and that's fine and that's often
2:50
good so use your plot judiciously to
2:53
tell a story and you may want to go from
2:55
one plot to the next to tell an
2:57
effective story rather than trying to
2:59
cram everything into a single plot so
3:01
now let's take a look at some examples
3:03
of what not to do when you're
3:05
visualizing data so here is a plot that
3:08
doesn't have a lot of information they
3:10
chose a reasonable default for x axes
3:13
and y axes so something that goes
3:16
between 0 and 1 and and that's often
3:18
very reasonable those could be
3:19
probabilities for instance but all the
3:22
data is down here in the corner are
3:26
there significant differences this Airy
3:28
trend in these data it's impossible to
3:30
tell and so because the axes are way too
3:34
large for what the actual data are it's
3:36
impossible to understand what's going on
3:39
here you can also air in the opposite
3:42
direction and so you might be cutting
3:47
your y-axis too much so that it looks
3:50
like there is a huge jump from point A
3:53
to point B when if you look at the y
3:56
axis there is a much smaller jump than
4:00
what you would believe if you just
4:01
looked at the size of these bars there
4:04
is a trade-off here and you can go too
4:07
far in either direction so try to be
4:09
reasonable and if you do truncate the y
4:12

axis where it doesn't start from 0 do so
4:14
in a way that is very clear and so make
4:19
sure that if you have a nonzero y axis
4:22
it's
4:23
at least eight that it's starting at
4:25
such-and-such value and if you're going
4:28
to make comparisons over time so for
4:29
example on the x-axis make sure that you
4:31
have enough points of comparison so for
4:34
this graph you may want to go further
4:36
back in time so like go back decades
4:40
like to the Reagan tax cut and show the
4:44
fluctuation of the tax rate over time to
4:47
show how this trend fits in with the
4:49
overall trend rather than just looking
4:51
at essentially two data points so take a
4:55
look at this graph and if you just
4:57
looked at the title of this
4:58
visualization you might believe that
5:01
this is trying to tell you a story about
5:04
how much various Governors make in
5:08
individual states one problem about this
5:13
though is that's not the easiest thing
5:15
to see what this visualization conveys
5:17
very clearly is that there are more
5:20
Republican governors and Democratic
5:22
governors and there's only one
5:23
independent governor that's interesting
5:27
but that's not what this visualization
5:29
is trying to convey it's trying to
5:31
convey something about salaries but to
5:34
understand the salaries you actually
5:36
have to read numbers and one diagnostic

5:41
for if you have a bad visualization is
5:44
if you threw in random information to
5:47
the thing that you care about would your
5:49
visualization look different and in this
5:51
case it wouldn't look different you
5:54
would still have exactly the same colors
5:56
exactly the same text breast showing
5:59
different numbers but if you have random
6:01
salary information
6:03
nothing would actually change so this
6:05
isn't allowing comparisons between
6:08
different states easily and so for
6:11
example let's say that I'm from Colorado
6:14
and I want to know who makes more than
6:18
Hickenlooper and who makes less than
6:20
Hickenlooper it's very difficult for me
6:22
to see that easily so if you did
6:25
something like shading the state based
6:28
on how much the governor was paid it
6:30
would be easier to see which are the
6:33
highly paid governor's and win
6:34
are the less well-paid Gunners so again
6:38
you want to enable comparisons and so
6:41
here you have a graphic that's trying to
6:44
show that trade between China and Taiwan
6:48
is different than trade between China
6:51
and the United States but what makes
6:54
this really difficult is that the colors
6:56
are inconsistent from one side to
6:58
another the y axes are not comparable
7:01
and so there is some valuable
7:05
information here but this visualization
7:08

doesn't help you get there
7:10
and so you need to translate the y axis
7:13
you need to flip the colors and then you
7:17
can understand what's going on if
7:19
anything this visualization hurts you in
7:22
trying to come to the right conclusion
7:24
about what's the difference between
7:25
trade between China and the US and trade
7:29
between Taiwan in the US so here's a
7:32
visualization that's trying to show how
7:34
various aspects of car properties
7:38
correlate with each other and here you
7:41
have things like the weight of the car
7:44
and the mileage mpg of a car are
7:50
negatively correlated the heavier your
7:53
car is the less mileage that it's going
7:56
to get you can see that by looking at
7:58
this cell here and so this makes sense
8:01
there's a negative correlation between
8:03
the weight of the car and the mileage
8:06
but it's really hard to see the
8:08
properties that correlate with each
8:11
other it's a mess of red and blue
8:14
throughout this entire visualization if
8:17
you instead order your properties in a
8:21
reasonable way so for example trying to
8:24
put the things that have the highest
8:25
correlations down here and things with
8:32
lowest correlations in these two corners
8:36
you have a much better visualization
8:38
that allows you to see what are the
8:41
things that are positively correlated
8:43
with each other so for example the

8:47
attorney
8:48
radius and the displacement are
8:53
positively correlated with each other
8:55
and things like weight and mpg are down
9:00
in this corner where you have poorer
9:03
correlations so again this is a place
9:05
where the default ordering for example a
9:07
medical order doesn't quite work but if
9:09
you do something more logical you can
9:11
tell a much clearer story to wrap up
9:13
often when you're working on a data
9:15
science project you will go through a
9:17
number of steps and this isn't really
9:20
sequential sometimes it's more of a
9:21
cycle you'll go back and forth but let's
9:24
treat it as an idealized sequence you
9:27
get some data you wrangle it so that you
9:30
get it into a clean format then you do a
9:32
visualization to explain to yourself
9:35
what's going on with the data is it
9:37
possible to build a model on this data
9:39
what are the ranges and limits of the
9:44
data that we have and then you build
9:46
some model and then you want to use it
9:48
for something in the world and part of
9:51
the process of using your data is
9:53
explaining to someone else how your
9:55
model works how it will work on typical
9:57
data and that again requires
9:59
visualization for you to do that so you
10:01
need visualization before you build the
10:03
model so that you yourself can
10:04

understand what's going on and then you
10:06
need visualization to tell the world
10:07
what's going on and why your model
10:09
worked well and then as I said often
10:14
this is a cycle and so once you have a
10:16
model you get new data and the process
10:18
starts from the very beginning one thing
10:21
that I want to emphasize is that copying
10:23
is okay and especially for
10:26
visualizations you'll see visualizations
10:28
that look great that do a good job of
10:30
explaining what's going on and it's
10:32
perfectly okay to copy that
10:35
give attribution show where you got it
10:38
from but imitation is the sincerest form
10:41
of flattery and you should use good
10:45
aspects of visualization in your own
10:47
work and if you see something that's
10:49
done really well figure out how to
10:51
replicate it with your own data and your
10:53
own tools so this was our very very
10:56
brief introduction to
10:57
nation will be playing around with it in
10:59
class and trying to interact with some
11:01
data and you'll be able to play around
11:04
with your own data and as we build
11:07
models we'll be using these sorts of
11:08
visualizations to see how well these
11:10
models are doing and to figure out ways
11:12
that we can improve our models